Lab No: 01

✵ Lab Title: Installation of C++ and Running First Program

- **Learning objectives:**

- *The objective of this lab is:*

- *To make a group of three students*
- *To install C++ compiler and Visual Studio Code*
- *To configure C++ in Visual Studio Code*
- *To run a simple C++ program*
- *To display the names of group members using C++*

Software Requirements

The following software is required:

- Visual Studio Code
- C++ Compiler (MinGW / GCC)
- Windows Operating System.

- *Theory:*

1. *C++ is a programming language used to develop software, applications, and systems.*
   Visual Studio Code (VS Code) is a code editor used to write and run C++ programs.
2. *To run C++ in VS Code, we must install:*

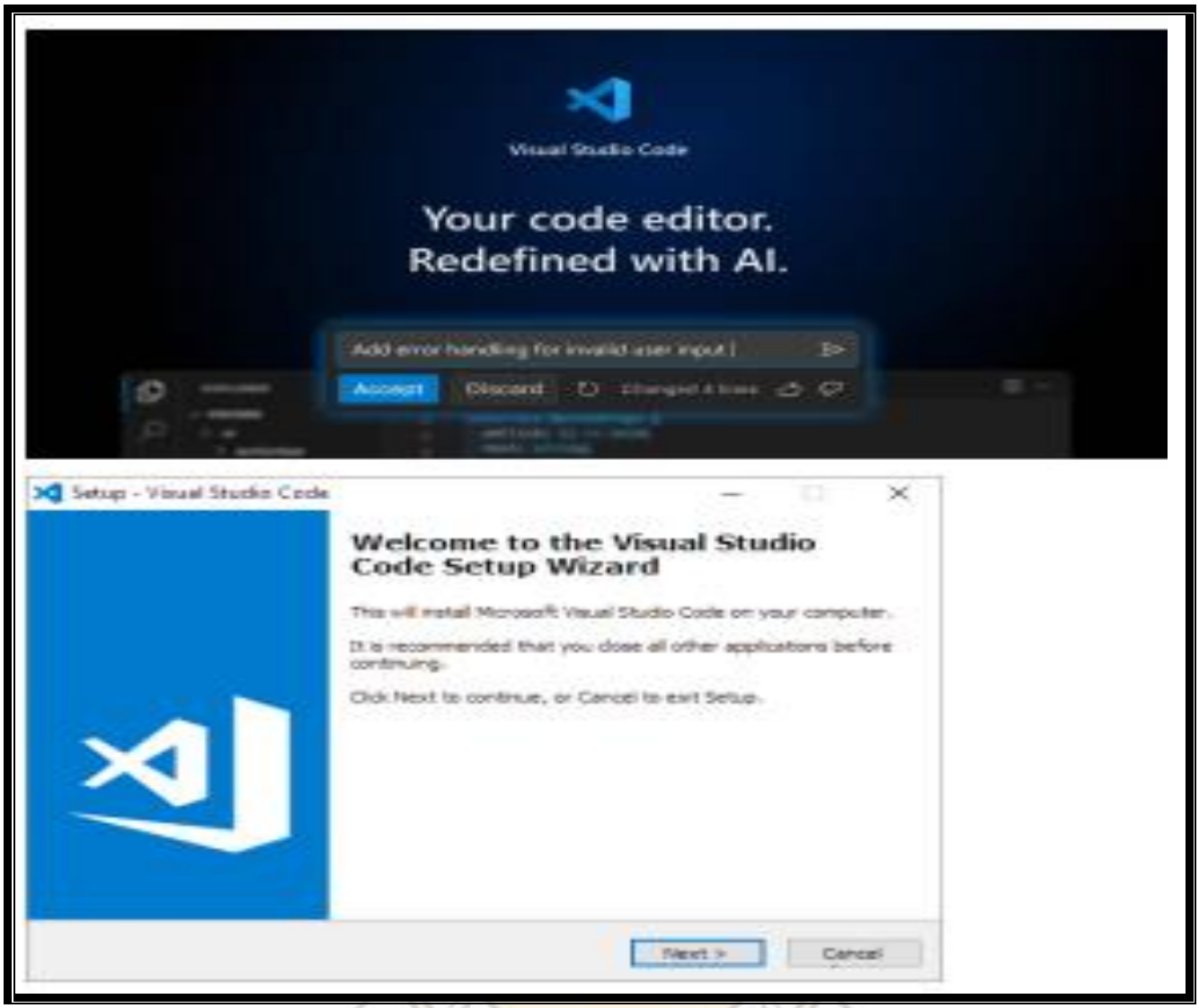
- *VS Code editor*
- *C++ compiler*
- *Required extensions*

- ***Procedure***

- *Step 1: Download Visual Studio Code*

1. *Open browser*
2. *Go to official website: https://code.visualstudio.com*
3. *Click Download*
4. *Install VS Code*

5. **Purpose:** Visual Studio Code is used to write and run C++ programs.



Step2:Installation of MinGW Compiler

steps:

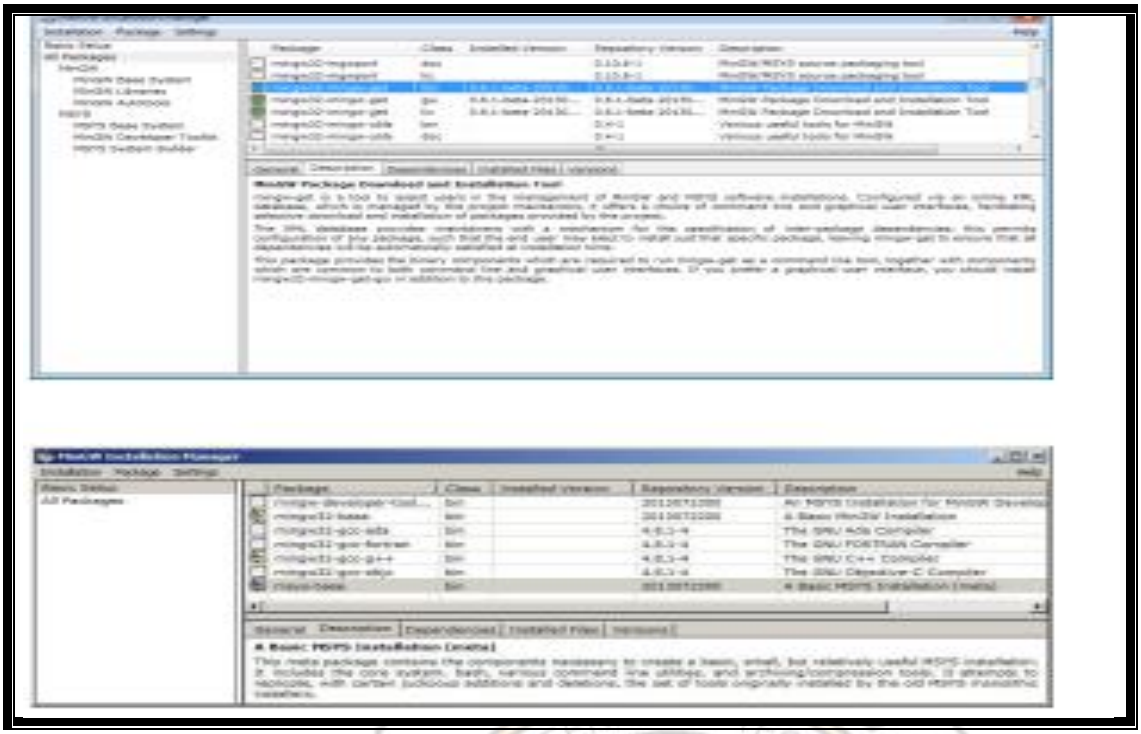
1. Opened website https://sourceforge.net/projects/mingw/

2. Downloaded MinGW installer

3. Installed MinGW software

4. Opened Installation Manager

5. Selected package:

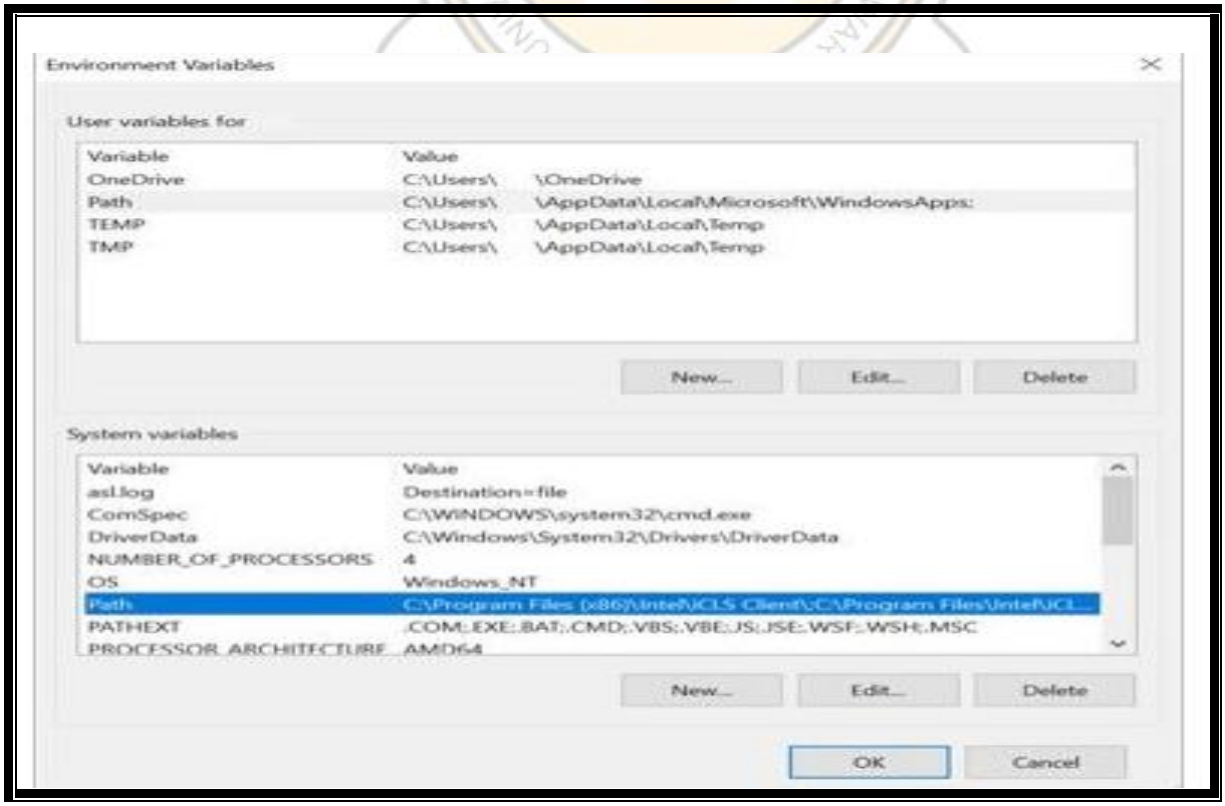


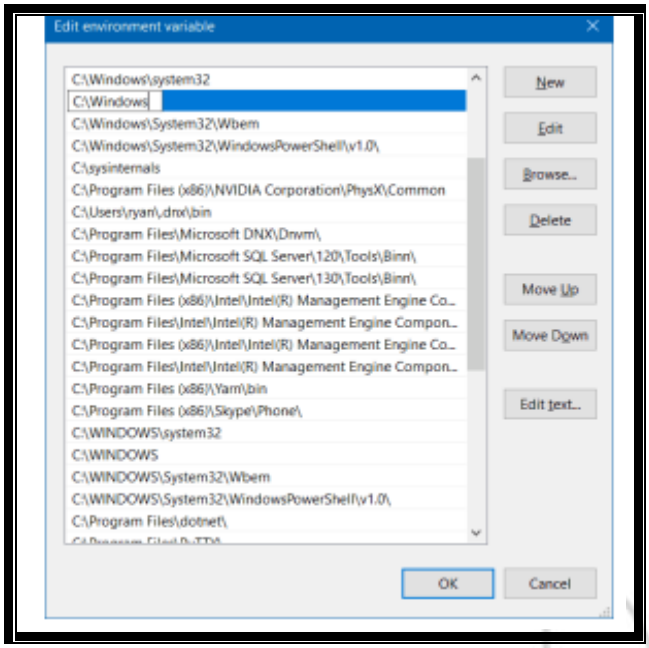
mingw32-gcc-g++

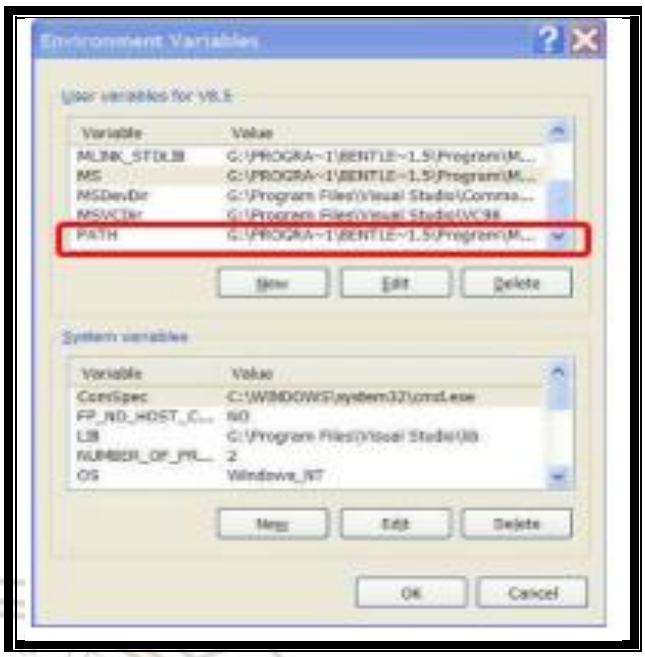
6. Clicked Install → Apply Changes

Purpose: MinGW is used to compile C++ programs.

Step 3.Adding MinGW Path to Environment Variables







Steps:

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1. Copied path: C:\MinGW\bin
2. Opened Environment Variables
3. Clicked Path → Edit
4. Clicked New
5. Added path
6. Clicked OK

Purpose: This allows system to recognize the g++ compiler.

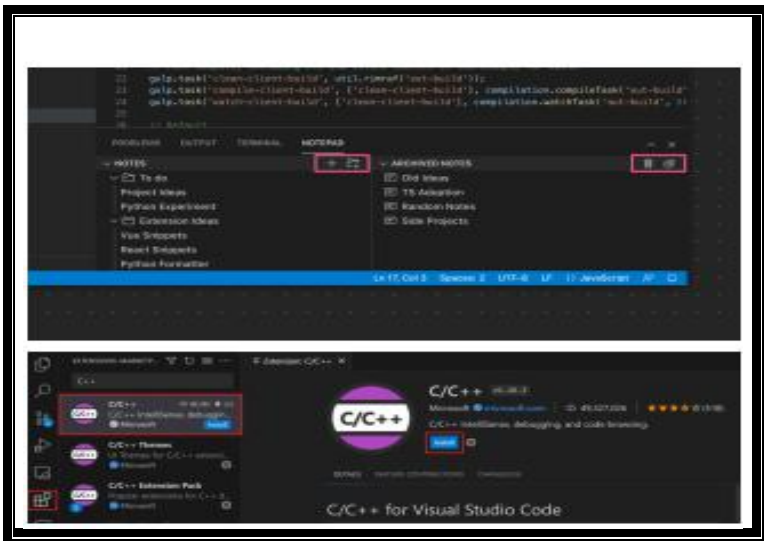
Step 4.installation of Extensions in Visual Studio Code

Extensions Installed:

• C/C++ Extension

• Code Runner Extension

Purpose: These extensions help in compiling and running C++ program.



Lab tasks :perform one by one all tasks

Task 1: Make group of three students

Group Members: 1. Hidayat Ullah 2. Anis Ahmad 3. Jawad Ali

Task 2: Understand lab instructions

All instructions were read and understood.

Task 3: Complete installation and setup

Successfully installed:

- Visual Studio Code
- MinGW Compiler
- Environment Variables
- Extensions

- **Task No: 04**
- **Write a simple C++program.**

```
#include <iostream>
using namespace std;

int main()
{
    cout << "Group Members:" << endl;

    cout << "Hidayat Ullah" << endl;
    cout << "Anis Ahmad" << endl;
    cout << "Jawad Ali" << endl;

    return 0;
}
```

Output:

Group Members:
Hidayat Ullah
Anis Ahmad
Jawad Ali

✍ Conclusion of lab:

✍ In this lab, we learned:
✍ • Installation of Visual Studio Code
✍ • Installation of MinGW Compiler
✍ • Setting Environment Variables
✍ • Installing Extensions
✍ • Running C++ Program
✍ The C++ development environment is now ready for programming.